

Enge Lou

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EDUCATION

Nanyang Technological University, Singapore

Aug 2025 - Dec 2026

Master of Science in Artificial Intelligence

- Relevant Modules: (1) Multi-Agent System (2) Generative AI (3) Time Series Analysis (4) Large Language Models

Nanyang Technological University, Singapore

Jul 2022 - Jun 2025

Bachelor of Engineering (Electrical & Electronic Engineering)

- **CGPA: 4.25/5.00**
- Area of specialization: *Data Analysis & Machine Learning (Computing & Intelligent systems)*
- Relevant Modules: (1) Machine Learning Design & Application (2) Artificial Intelligence & Data Mining

INTERNSHIP EXPERIENCES

Lobah Play, Singapore

Aug 2026 – Present

AI Research Engineer Intern

- Translated requirements from the business team and end users into mobile game deliverables, owning each title from spec to delivery.
- Built "Dark Factory", an agent-driven game production pipeline that raised throughput from 4–5 games per week to 4–5 games every 2–3 days; currently extending it toward unattended 24/7 operation.

Vector Future, China

Apr 2026 - Jul 2026

Forward Deployed AI Engineer Intern (Part-time, Project-based)

- Worked with client teams in an iterative co-delivery model to scope requirements, prototype workflows, and deliver an AI customer-operations platform through production handover.
- Translated the client's 15-year operating methodology into AI workflows for customer profiling, intent detection, product recommendations, after-sales support, and human escalation; integrated WeChat, CRM, order, logistics, coupon, and internal data systems.
- Deployed the platform for a customer-operations team of approximately 50 staff, supporting 7,000–20,000 daily messages and reducing average customer-service response time by 60%.

Desay SV Automotive, Singapore

Jan 2026 - May 2026

LLM Applications Engineer Intern

- Built and deployed an internal multilingual translation plugin supporting text and PDF formats across Chinese, English, Japanese, Spanish and German; leveraged **locally-deployed LLMs** with a RAG-enhanced knowledge base to ensure terminology consistency, supporting cross-regional engineering collaboration.
- Built a PRD–UX–UI consistency audit system in partnership with the product and design departments: co-ran requirement workshops and progress reviews with both teams, evaluated OpenCV, YOLOv8, OCR and LLM approaches, and delivered a pilot reaching 95% inconsistency detection with a roadmap for automated QA.
- Co-authored *Structured Runtime Safe Monitoring for Camera-Based Driver Monitoring Systems*, accepted at IEEE-ITSC 2026: designed the core algorithm framework, ran experiments and co-wrote the manuscript.
- Integrated OpenClaw with ROS2 and Gazebo to prototype LLM-driven robotic-arm control, enabling the LLM to issue motion commands from natural-language instructions; validated the first-phase command loop in simulation.
- Built a scheduled LLM news-digest service to automate internal technology-trend tracking.

A*STAR CFAR, Singapore

Aug 2025 - Jan 2026

Junior Scientist I

- Built PyTorch pipelines using ESM protein language models for enzyme-activity prediction on large, imbalanced datasets.
- Automated 3D structure generation with RhoFold and API-based sequence retrieval to construct a 400k+ record dataset for RNA–ligand binding prediction; maintained reproducible preprocessing, training and inference workflows.

PROJECTS & RESEARCH

ISolid — AI-Powered Idea Structuring System

May 2026 - Aug 2026

- Built and shipped a full-stack agent product that turns vague ideas into production-ready PRDs, GTD lists and automated accountability pings; ran a closed beta with 50 early-access users, focused on structured-output reliability.
- Architected a three-graph **LangGraph** system (intake → finalize → monitor) featuring state isolation and dynamic node routing; implemented a tiered LLM dispatch layer (leveraging **Claude Sonnet 4.6 & GPT-5.5**) that optimizes for reasoning quality and cost-efficiency.
- Engineered a multi-layer Pydantic validation system to unify structured data across different model providers, enabling zero-crash model fallback; enforced rigid prompt injection defenses at the system message level across all inference sites.
- Deployed on VPS via **Docker Compose + Nginx**; configured specialized reverse proxy settings to handle long-duration (30–300s) LLM streaming, and implemented a lightweight i18n system with server-side persistence for global scalability.

On-Chain Automated Trading Engine

Jul 2026 - Present

- Architected an autonomous, end-to-end trading system driven by 3 asynchronous background daemons (Signal → Execution → Settlement → Audit); managed 26 active positions across multiple accounts with an 8-second end-to-end signal latency.
- Built a multi-layer risk framework filtering every signal through 5 guards, achieving a 62% directional win rate on early settled positions.
- Engineered a fault-tolerant execution layer with multi-node RPC failover, achieving zero downtime; built an independent P&L reconciliation layer to recompute equity directly from on-chain state, automating position settlement and redemption.

ShieldFlow – Multi-Agent Data Analysis Engine

Aug 2025 - Jan 2026

- Architected an autonomous multi-agent analytical engine (Coordinator-Analyst-Verifier) that converts natural language into interactive Plotly visualizations, with a self-correction loop that automatically refines generated code on execution errors.
- Engineered a secure, reliable execution stack: AST-based static auditing with a network-isolated Docker Sandbox to mitigate arbitrary code execution, a Schema RAG pipeline (ChromaDB) injecting column metadata to reduce mapping errors, and an SQLite persistence layer for multi-turn memory and cost observability.

SKILLS

Languages: Proficient in English and Chinese

Technical Skills: Python (LangGraph, LiteLLM, FastAPI, PyTorch), SQL, Java, Docker Compose, Nginx, Linux

Agentic AI & LLMs: Multi-Agent Workflows, Schema RAG, Vector Embeddings (ChromaDB), Structured Output & Validation, LLM Evals, Prompt Injection Defence, Cost & Latency Observability

AI Methodologies: Time Series Analysis, Computer Vision (YOLOv8, OpenCV), Self-Attention Mechanisms, Transfer Learning, Data Augmentation